MIL-R-81435A(AS)
15 March 1969
SUPERSEDING
MIL-R-81435(WP)
1 APRIL 1966

MILITARY SPECIFICATION

Receiver-Transmitter, Radio RT-859( )/APX-72
and associated
Mountings MT-3809( )/APX-72 and MT-3948( )/APX-72

This specification has been approved by the
Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall provide automatic radar identification of an airborne or surface vehicle to all suitably-equipped challenging aircraft, ships and ground forces within the operational range of the system.

1.2 Classification - The equipment covered by this specification will consist of the items listed in table I.

Table I. Equipment Classification

Type Designation	Applicable Paragraph
Receiver-Transmitter, Radio RT-859( )/APX-72	3.5.1
Mounting MT-3809( )/APX-72	3.5.3.1
Mounting MT-3948( )/APX-72	3.5.3.2

1.3 Associated Equipment - The equipment will operate with the associated equipment listed in 6.8.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents form a part of this specification to the extent specified herein. Listed are the issues of the documents in effect for the previous procurement. However, in lieu

FSC 5821

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of the issue listed, the contractor shall use the latest issue in effect of these documents where feasible. If the use of the latest issue will affect design, performance, or interchangeability of any replaceable part, then the issue of the document listed below shall be used.

SPECIFICATIONS

Military

MIL-W-5088C 26 May 1965	Wiring; Aircraft, Installation of
MIL-E-5400H 1 June 1965	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422E 13 April 1959 And Amendment 2 15 November 1961	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181D 25 November 1959	Interference Control Requirements, Aircraft Equipment
MIL-E-17555E 26 July 1960	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303A 1 May 1963	Test Procedures; Preproduction and Acceptance for Aircraft Electronic Equipment, Format for
MIL-N-18307C 1 October 1958 And Amendment 3 6 August 1963	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-S-23603 1 March 1963	System Readiness/Maintainability; Avionic Systems Design, General Specification for
MIL-C-38257 31 March 1964	Control, Transponder Set, C-6280( )(P)/APX
MIL-E-55301 1 April 1965 And Amendment 1 1 March 1966	Electromagnetic Compatibility

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MIL-I-81219 1 June 1965 And Supplement-1 1 June 1965	Indicator, Elapsed Time, Electrochemical

Naval Air Systems Command

WR-98 15 December 1965	Microelectronic Devices Used in Avionics Equipment, Procedures for Selection and Approval of

STANDARDS

Military

MIL-STD-704 6 October 1959	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781A 10 December 1965	Reliability Tests, Exponential Distribution
MIL-STD-785 30 June 1965	Reliability Program (for Systems and equipments), Requirements for
MS-90386-4 1 September 1965	Indicator, Elapsed Time, Electrochemical (Mercury) 28 Volt DC
MIL-STD-794 10 March 1965	Parts and Equipment, Procedures for Packaging and Packing of

2.1.1 Availability of Documents - When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

3.2 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

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a. Microelectronic technology shall be considered and microelectronic items shall conform to requirements specified herein.

b. Other parts and materials requirements shall conform to MIL-E-5400.

c. Nonreparable subassemblies, as outlined in MIL-E-5400, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonreparable subassemblies must be reliable. (See 6.5).

d. When previously produced models of this equipment did not use nonreparable subassemblies, the design shall not be changed to employ nonreparable assemblies without the approval of the procuring activity.

e. All plug-in boards described in 3.5.1.4.3.1 through 3.5.1.4.3.7 shall have means of self locking and extraction from the main assembly. This means of locking and extracting shall be approved by the procuring activity.

3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials (including electron tubes, transistors and diodes) other than microelectronic devices, shall be obtained as specified in MIL-E-5400. Microelectronic devices shall be approved in accordance with WR-98.

3.2.2 Microelectronic Modular Assemblies - When used, microelectronic modular assemblies shall meet the requirements of WR-98.

3.2.3 Modules - The electronic portions of the equipment shall be divided into maintenance modules in accordance with MIL-S-23603. Maintenance modules shall normally be considered reparable.

3.3 Design and Construction - The equipment shall conform with all the applicable requirements of MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

*3.3.1 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 16.5 pounds including the mount specified in 3.5.3.1 nor exceed 17.15 pounds including the mount specified in 3.5.3.2.

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3.3.2 Reliability - The contractor shall conduct a reliability program using MIL-STD-785 as a guide.

3.3.2.1 Operational Stability - The equipment shall operate with satisfactory performance (3.4), continuously or intermittently for a period of at least 500 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating Life - The equipment shall have a total operating life, as defined in MIL-STD-781, of 5000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

*3.3.2.3 Reliability in Mean Time Between Failures (MTBF) - The equipment shall have a specified mean time between failures of 300 hours when tested and accepted as outlined under the requirements of 4.4.3.

3.3.2.4 Time Totalizing Meter - The RT-849( )/APX-72 shall contain a time totalizing meter in accordance with MIL-I-81219 and MS-90386-4, except that the hour identification scale shall be rotated 90° so as to be read from the horizontal position.

3.3.3 Cabling and Connections -

3.3.3.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with MIL-E-5400.

3.3.3.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of MIL-W-5088. The external wiring shall be unshielded except when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.3.4 Interchangeability - The equipment shall meet the interchangeability requirements of MIL-E-5400.

3.3.5 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of MIL-I-6181 and MIL-E-55301. Where a conflict of requirements exists, the more stringent requirements shall take precedence.

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3.3.6 Provisions for Maintainability - Provisions for maintenance shall be as specified in MIL-E-5400. Specific test points and test facilites shall be provided to the greatest extent practicable for ease of field testing and maintenace.

*3.3.7 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with MIL-N-18307. Mountings (3.5.3) shall have nameplates in accordance with the foregoing except that attachment of the plates may be made by means of an approved adhesive in lieu of screws.

3.3.8 Standard Conditions - The conditions listed in Table II shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Table II - Standard Conditions for Laboratory Tests

Temperature	Room ambient (25°C ± 5°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90 percent relative humidity
Input power voltage	115 ± 1.0 VAC and 27.5 ± 0.5 VDC
Interrogation Rate	500 interrogations/ second when not otherwise specified or required.

3.3.9 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-E-5400 for Class 2 equipment, except as modified herein.

3.3.9.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curve IV of MIL-E-5400. with isolators removed.

3.3.9.2 Temperature - Altitude Combination - The equipment shall operate continuously under the temperature - altitude combinations of Curve B, Figure 4 of MIL-E-5400 for Class 2 equipment.

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3.3.9.3 Antenna Mismatch - Anticipated service conditions of impedance mismatch shall include all values of load impedance that present a voltage standing wave ratio (SWR) up to and including 3.5 decibels (DB) (1.497:1 SWR) of any phase relationship at the antenna terminal of the receiver-transmitter unit.

3.3.9.4 Transmitter Duty Cycle - The transmitter shall be capable of continuous operation at 1.0 percent duty cycle. For mean-timebetween failure (MTBF) and life-test considerations, operations shall be at 0.5 percent duty-cycle for 50 minutes and 1.0 percent for 10 minutes of each hour. For preproduction and acceptance testing, normal duty-cycle shall be 0.5 percent and limit conditions shall 0.1 percent and 1.0 percent duty-cycles. Duty cycle is defined in 6.9.1.

*3.3.10 Warm-up Time - The time required for the equipment to warm-up prior to operation shall be kept to a minimum and shall not exceed two minutes under either standard or extreme service conditions.

*3.3.11 Input Electrical Power -

*3.3.11.1 Operating Power - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources with characteristics as defined in MIL-STD-704 having limits as specified therein. The power required shall not exceed the specified amounts at transponder duty cycles from 0 to 1.0 percent. AC or DC operation shall be selectable by a wiring jumper in the aircraft wiring to the front panel connector.

a. AC Power (Single Phase) 115 volts (V), 400 Hz, Category "B" 100 volt amperes (VA) (90 watts) exclusive of switched input power (3.3.12.4). The maximum surge current shall not exceed 10 amperes.

b. DC Power 28V Category "B", 75 watts, (over the input voltage range of 17 to 29V) exclusive of switched input power (3.3.11.4) and switched current from caution and reply lights (3.4.2.4).

c. When the input power is alternating current (AC), direct current (DC) power (28 V) is required and shall not exceed 0.30 amperes for the operation of internal relays.

Power supply current per 3.4.6.1 are excluded.

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3.3.11.2 MIL-STD-704 Utilization Checklist - The considerations checked in Figure 1 indicate areas where provisions of MIL-STD-704 are applicable.

Item	No	Yes	Note
1. MIL-STD-704 POWER REQUIRED (AC)		X	
2. MIL-STD-704 POWER REQUIRED (DC)		X	
3. OTHER TYPE POWER REQUIRED	X		
4. CATEGORY "A" EQUIPMENT			
5. CATEGORY "B" EQUIPMENT		X	
6. CATEGORY "C" EQUIPMENT			
7. DURING NORMAL OPERATION REGIONS AND DEGREES OF DEGRADED PERFORMANCE ARE PERMITTED	X		
8. DURING ABNORMAL OPERATION, REGIONS AND DEGREES OF PERFORMANCE ARE REQUIRED	X		
9. MANUAL RESET IS PERMITTED AFTER ABNORMAL OPERATION	X		
10. FULL EQUIPMENT PERFORMANCE IS REQUIRED FOR THE FOLLOWING AIRCRAFT OPERATING CONDITIONS:			
START AND WARMUP		X	
TAKEOFF AND CLIMB		X	
CRUISE AND CRUISE-COMBAT		X	
LANDING		X	
EMERGENCY	X		#
11. EQUIPMENT PERFORMANCE IS REQUIRED WITH LOSS OF POWER FROM ONE PHASE	X		

See 3.3.11.3, 3.4.1.1, 3.4.1.8, and 3.4.5.3.1.

Figure 1. MIL-STD-704 Utilization Checklist

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3.3.11.3 Degraded Performance - Degraded performance will be permitted for voltage transients of time duration not exceeding 0.5 second during normal AC or DC electric system operation. Operation shall return to normal with no resulting damage to the equipment.

3.3.11.4 Switched Input Power - When energized and placed in any operating position except OFF, the equipment shall interconnect input power to jack J1 as specified in 3.5.1.6.1 The 28 VDC switched output shall be rated for 2.0 amperes maximum. The 115 VAC switched output shall be rated for 1.0 ampere maximum. The switched DC and AC outputs shall not be fused.

3.3.11.5 AC Line Balance - Both sides of the AC line shall be balanced with respect to ground within the equipment.

3.3.12 Cooling - The equipment shall not require either an internal blower or external cooling air.

3.3.13 Interrogation Characteristics - The equipment shall accept interrogation signals with characteristics in accordance with tables III and IV:

Table III. Interrogation Mode Spacing

Mode	Interrogation Characteristic
1	2 pulses spaced 3 ± 0.2 microseconds (USEC)
2	2 pulses spaced 5 ± 0.2 USEC
3/A	2 pulses spaced 8 ± 0.2 USEC
Test	2 pulses spaced 6.5 ± 0.2 USEC
4	4 pulses referenced from the first pulse in multiples of 2 USEC. Each pulse should be located within ± 0.1 USEC from the reference position. The 4 pulses may be followed by as many as 33 additional pulses, spaced as close as 2.0 USEC.
C	2 pulses spaced 21.0 ± 0.2 USEC

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Table IV. Input Pulse Characteristics

Mode	Interrogation Code	
1,2,3/A, C, and Test	Pulse Width	0.8 ± 0.1 USEC
	Rise Time	0.05 to 0.1 USEC
	Decay Time	0.05 to 0.2 USEC
4	Pulse Width	0.5 ± 0.1 USEC
	Rise Time	0.05 to 0.1 USEC
	Decay Time	0.05 to 0.2 USEC

3.3.14 Input Protection - The equipment shall not be damaged by external signals of any frequency or type of modulation of up to 10 V peak amplitude and average power of -20 DBM applied directly to the antenna terminal.

3.4 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service and input power conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.4.1 Operation - The equipment shall receive coded interrogations (3.3.13), shall decode these interrogations, and in response, shall transmit proper pulse replies.

3.4.1.1 Input Voltage Range -

a. AC - The equipment shall give full specified performance when operated with steady-state input voltages between 107 and 122 VAC. The equipment shall function in the ranges of 103 to 107 and 122 to 127 VAC but performance may be degraded, no damage to components shall occur, and full performance shall be restored when the input voltage returns to normal.

b. DC - The equipment shall give full specified performance when operated with steady-state input voltage from 21 to 29 VDC. The equipment shall function in the range of 17 to 21 VDC. Full specified performance shall be restored when the input voltage returns to normal. Relays shall function properly down to and including 17 VDC.

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3.4.1.2 Interference Characteristics - The equipment shall operate in the presence of the following types of interfering signals with minimum degradation:

3.4.1.2.1 Continuous Wave and Amplitude Modulated Continuous Wave Interference - The equipment shall operate in the presence of interfering signals consisting of continuous wave (CW) and amplitude-modulated continuous wave (AMCW) and shall meet the following requirements. Measurements shall be made to assure compliance throughout the required range of interrogating signal levels.

a. If the interrogating signal to which the equipment should reply is between 3 and 50 DB above the normal triggering level, an interfering signal 10 DB below the interrogating signal shall not reduce the number of valid replies more than 50 percent. At each level tested the interfering signal frequency shall be adjusted for maximum countdown of replies.

b. The AMCW signal shall be simulated using a signal generator modulated 30 percent by a 1000 Hertz (Hz) sine wave.

c. If the interrogating signal is between 20 and 50 DB above the normal triggering level, an interfering signal at a level 3 DB down from the interrogating signal shall not reduce the number of replies by more than 50 percent, except when the interfering signal combines with the interrogating signal in such a way that the amplitude modulation of the resultant signal is inverted or eliminated.

d. In the presence of CW and AMCW interference as specified herein, the equipment shall not produce more than 10 random or unsynchronized replies per second.

3.4.1.2.2 Pulse Interference - The receiver shall operate in the presence of interfering signals consisting of randomly spaced pulses and regularly spaced pulses and shall meet the following requirements:

a. If the interrogating signal to which the equipment should reply is between 3 and 10 DB above the normal triggering level, an interfering signal 10 DB down from the interrogating signal shall not reduce the number of replies by more than 50 percent.

b. If the interrogating signal is between 20 and 50 DB above the normal triggering level, an interfering signal 3 DB down from the interrogating signal shall not reduce the number of replies by more than 50 percent.

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c. If the interfering signal consists of randomly-spaced pulses only, and the interrogating signal is a Mode 1,2,3/A or C interrogation that is 10 DB above the interfering signal, the equipment shall reply to at least 90 percent of the interrogations where the randomly spaced pulses are not recognized as normal or side-lobe interrogations.

#3.4.1.3 Desensitization - Upon receipt of any pulse more than 0.7 USEC in duration, the receiver shall be desensitized by an amount that is within at least 9 DB of the amplitude of the desensitizing pulse, but shall at no time exceed the amplitude of the desensitizing pulse, with the exception of possible over-shoot during the first USEC following the desensitizing pulse.

3.4.1.4 Mode 4 Desensitization - Upon receipt of a group of four synchronization pulses (each more than 0.4 USEC in duration and spaced by 2 USEC) the sensitivity of the receiver shall be lowered so that an additional pulse 9 DB or more below the signal level of the synchronization pulse group should not be detected by the Mode 4 decoder. Upon receipt of a group of four pulses (each more than 0.4 USEC in duration and spaced by 4 USEC) the sensitivity of the receiver shall be lowered so that an additional pulse 12 DB or more below the signal level of the four pulse group and spaced 2 USEC after the last pulse of the four pulse group should not be detected by the Mode 4 decoder. With the exception of the first USEC following the trailing edge of the last desensitizing pulse, the sensitivity of the receiver shall be such that an additional pulse at the same signal level as either of the pulse groups will be detected by the Mode 4 decoder.

3.4.1.5 Recovery - Following desensitization, the receiver shall recover sensitivity (within 3 DB of minimum triggering level) within 15 USEC after reception of a desensitizing pulse having a signal strength up to 50 DB above the minimum triggering level (MTL).

#3.4.1.6 Minimum Triggering Level - Minimum triggering level is that level of signal required for 90 percent replies. The MTL of the equipment at the receiver antenna jack, when measured with nominal interrogation modes and at a frequency of 1030 ± 0.2 MHz under standard conditions(3.3.8) shall be at least -90 decibels referred to one volt (DBV) (-77 decibels referred to one milliwatt into 50 ohms (DBM) for all modes.) The total variation of MTL between Modes 1, 2, 3/A, C and TEST shall not exceed 1.0 DB when measured as above.

Single pulses of duration less than 0.7 USEC are not required to cause the specified desensitization, and should not cause desensitization greater than permitted by 3.4.1.3 and 3.4.1.4.

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The total variation of MTL between Mode 4 and the other Modes shall not exceed 2.0 DB when measured as above. Also, the MTL of the equipment on Modes 1, 2, 3/A, C and TEST shall not change by more than ±1.0 DB or by more than ±2.0 DB in Mode 4 when the interrogation pulse width is varied over the tolerance specified in 3.3.13 for each mode.

*3.4.1.7 Triggering Level Drift - The triggering level, when set at -90 DBV (-77 DBM) under standard conditions (3.3.8) shall not vary by more than ±3 DB over all combinations of service conditions.

3.4.1.8 Reduced Triggering Level Performance - When the power supply is operating on the direct-current source, MTL as measured under conditions of 3.3.8 may be reduced no more than 6.0 DB when the input voltage is reduced to 17.0 VDC.

3.4.1.9 Random Triggering Rate - The equipment shall not transmit more than 5 pulse trains per second averaged over one minute under the following conditions:

a. The receiver antenna receptacle is terminated in 51-ohm resistive load.

b. No external signal is present.

c. Triggering level (sensitivity) control is set for an MTL of at least -90 DBV (-77 DBM) as required in 3.4.1.6.

3.4.1.10 Triggering Level Selection - Provisions shall be included for remote selection between two triggering levels as follows:

a. Normal triggering level internally adjustable from at least -82 DBV (-69 DBM) to -90 DBV (-77 DBM).

b. Low triggering level internally adjustable from at least -80 DBV (-67 DBM) to -65 DBV (-52 DBM).

3.4.2 Overall Performance Requirements -

3.4.2.1 Automatic Overload Control - The automatic overload control (AOC) shall operate, by means of trigger level-reduction, to perform the following functions:

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3.4.2.1.1 Reply Duty-Cycle Control - The reply duty-cycle shall be controlled to protect the transmitter. The AOC action shall be equally effective in all modes. No more than 3 DB reduction in sensitivity shall occur at 0.9 percent duty-cycle. Maximum duty-cycle shall not exceed 1.1 percent when interrogation signals are at 50 DB above minimum triggering level.

3.4.2.1.2 Reply Rate Limiting - This control shall limit the number of generated reply trains in Modes 1, 2, 3/A, C, and Test, without regard to the number of pulses in each reply, or whether replies are Normal, Identification of Position (IDENT) or Emergency. Trigger level reduction shall be gradual to prevent weaker interrogations from denying transponder service to stronger interrogations as a result of over-interrogation. Maximum reply rate shall not exceed 120 percent of the rate at which reduction first begins if the interrogation signal is increased to 50 DB above MTL. Maximum reply rate shall be adjustable from 500 to 3000 replies per second, with interrogation signals 3 DB above MTL, and shall be set to 1200 replies per second prior to delivery.

3.4.2.1.3 Mode 4 Duty Limiter - This control shall limit the rate of decoding Mode 4 interrogation sync patterns by gradual reduction of triggering level. The threshold for reduction shall be adjustable between 1500 and 2500 Mode 4 enabling triggers per second with interrogation signals 3 DB above MTL. The maximum rate shall not exceed 120 percent of the rate at which reduction first begins if the interrogation signal is increased to 50 DB above MTL. When disparity pulses are present between 4 and 74 USEC following the fourth sync pulse, the limit threshold shall be increased by an amount inversely proportional to the spacing of the disparity pulse from the fourth sync pulse.

3.4.2.1.4 Side-Lobe Suppression Rate Limiter - This control shall limit the rate of side-lobe decoding on Modes 1, 2, 3/A and C by gradual reduction of triggering level. At a side-lobe suppression rate up to 5000 pulses per second there shall be no effect on the triggering level. With an interrogation signal 50 DB above MTL, maximum side lobe suppression rate shall not exceed 10,000 pulses per second.

3.4.2.1.5 Automatic Overload Control Stability - The absolute limits stated in 3.4.2.1.1 and 3.4.2.1.4 shall apply under all service conditions. The rates established by adjustment in 3.4.2.1.2 and 3.4.2.1.3 under conditons of 3.3.8 shall hold within ± 20 percent under all service conditions.

3.4.2.1.6 Reply Rate Limiting Action and Recovery Time - If the interrogation rate is suddenly increased from 1000 to 2500 pulse pairs per second when the threshold established by the control described in 3.4.2.1.2 has been adjusted to 2000 replies per second, the sensitivity of the receiver shall be reduced by 20 DB, for interrogation pairs 20 DB above MTL, within 0.2 seconds. Upon decreasing the interrogation rate to 1000 pulse pairs per second the receiver shall return to within 3 DB of normal sensitivity within 0.025 seconds.

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3.4.2.2 Suppression Characteristics -

*3.4.2.1 Suppression During Mode 4 Interrogation - Decoding of Modes 1, 2, 3/A and C and Test shallbe suppressed when the first four pulses of a Mode 4 interrogation are received. Once a Mode 4 trigger has been generated, the generation of additional Mode 4 triggers shall likewise be suppressed. Suppression due to a Mode 4 interrogation shall continue for 75 to 125 USEC after the fourth pulse, unless this suppression is terminated by a disparity pulse. If a disparity pulse (3.4.2.15.1.2) is received during this suppression period, suppression due to the Mode 4 interrogation shall be ended within 2 USEC after receipt of the disparity pulse. The first two pulses of a Mode 4 interrogation shall also be recognized as a legitimate interrogation side lobe suppression (ISLS) control signal and be processed as required under 3.4.2.5. The ISLS suppression gate resulting from a Mode 4 interrogation may (but shall not necessarily) be reset by a disparity pulse coincident with it.

3.4.2.2.2 Suppression During Transmission of Mode 4 Replies - Decoding of Modes 1, 2, 3/A, 4, C and Test shall be suppressed during transmission of Mode 4 replies. Recovery to within 3 DB of minimum triggering level shall occur within 5.0 USEC after the last transmitted pulse.

3.4.2.2.3 Suppression During Transmission of Replies in Modes 1, 2, 3/A, C and Test - Decoding of Modes 1, 2, 3/A, 4, C, and Test shall be suppressed during transmission of replies in Modes 1, 2, 3/A, C and Test, Suppression shall continue for at least 75 USEC after the second transmitted framing pulse of a normal, IDENT or Emergency reply, but no longer than 100 USEC.

3.4.2.2.4 Suppression Input - Decoding of all modes shall be suppressed upon reception of a suppression pulse having characteristics as listed in table VI.

Table VI. Suppression Input Pulse Characteristics

Characteristic	Value
Polarity	Positive
Amplitude	15 to 70 V
Duration	1.5 to 250 USEC

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Input Impedance	2200 ± 10 percent ohms resistance and 50 picofarad (PF) or less capacitance
Rise Time (slope)	10V/USEC or greater
Decay Time (slope)	10V/USEC or greater
Noise	No suppression shall result from any input of less than 5.0V peak amplitude.

3.4.2.2.4.1 Suppression Operation - Upon application to the receiver-transmitter of a suppression pulse having the above characteristics, decoding of Modes 1, 2, 3/A, C, Test and 4 shall be suppressed for an interval starting not more than 1.0 USEC after the suppression pulse has reached an amplitude of 10 V and shall continue for at least the duration of the suppression pulse. After the end of the suppression pulse the receiver shall regain normal sensitivity, within 3 DB, no later than 15 USEC after the suppression pulse has fallen to 5.0V. The receiver-transmitter shall be insensitive to proper radio frequency (RF) interrogations on all modes, with signal levels of -20 DBV, for the duration of the suppression pulse. An input connection which is part of the main chassis connector shall be provided for this input.

3.4.2.2.5 External Suppression Output - The equipment shall generate a suppression pulse during the time of transmission for each Mode 1, 2, 3/A, C, and 4 output train, which is used to suppress other equipments. The suppression pulse shall have the characteristics listed in table VII.

Table VII. Suppression Output Pulse Characteristics

Characteristic	Value
Amplitude	18 to 30 V
Polarity	Positive
Synchronization	The pulse shall reach 18 V minimum by the time an RF output pulse has reached 10 percent of its detected amplitude

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Table (Cont.)

Duration	The suppression pulse shall remain at 18V or greater until the trailing edge of the last transmitted pulse has decayed to the 10 percent level. Suppression output shall also be less than 5V, 5.0 USEC after the last transmitted pulse of the reply train has fallen to 10 percent level.
Rise Time	20V or more per USEC
Decay Time	10V or more per USEC
System Impedance	The load impedance presented to the suppressor output shall not be more than 2200 ohms nor less than 300 ohms shunted by not more than 1850PF capacitance
Spurious Output	Combined steady-state and transient voltages shall not exceed 1.0V, either polarity, with the system loads described above.

3.4.2.3 Echo Rejection - The transponder shall contain echo rejection circuitry designed to permit normal operation in the presence of echoes of signals in space and be compatible with the requirements specified in 3.4.2.5.

3.4.2.4 Auxiliary DC Power Utilization - When the equipment is operating on either 115 VAC or 28 VDC primary input power, 28 VDC from the aircraft supply shall be utilized by the equipment for the operation of internal relays, auxiliary equipment and external lights as specified herein:

a. Provision shall be made to enable an external CAUTION light from the external 28 VDC input to the equipment. The equipment shall be capable of furnishing one ampere of current to the CAUTION light output lead.

b. The power for an external Mode 4 REPLY light shall be provided from the external 28 VDC input to the equipment. When the REPLY light is enabled the equipment shall be capable of furnishing one hundred milliamperes of current to the REPLY light output lead with a maximum internal voltage drop of 2 VDC under any combination of service conditions.

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c. The power for operation of relays internal to the equipment shall be obtained from the external 28 VDC input to the equipment.

3.4.2.5 Interrogation Side Lobe Suppression (ISLS)

3.4.2.5.1 ISLS Control Pulse Characteristics - The ISLS control pulse will have characteristics as listed in table VIII.

Table VIII. ISLS Control Pulse Characteristics

Characteristics	Value
Pulse Width	0.8 ± 0.1 USEC for Modes 1, 2, 3/A, C and Test, or 0.5 ± 0.1 USEC for Mode 4
Pulse Position	2 ± 0.15 USEC after the first pulse of a Mode 1, 2, 3/A, C and Test interrogation. 8.0 ± 0.15 USEC after the first pulse of a train of four 0.5 USEC pulses spaced 2 USEC for Mode 4
Rise Time	Between 0.05 and 0.1 USEC
Decay Time	Between 0.05 and 0.2 USEC
Frequency	Within ± 0.2 megahertz (MHz) of the frequency of the interrogation pulses (1030 MHz nominal)

3.4.2.5.2 Conditions For ISLS - The decoding of Modes 1, 2, 3/A, C, or Test shall be suppressed (not less than 99 percent efficiency for interrogations 3 DB or more above MTL) when the received amplitude of the ISLS control pulse is equal to or in excess of the received amplitude of the first interrogation pulse and spaced 2 ± 0.15 USEC from it. The suppression of the decoder shall be for a period of 35 ± 10 USEC and shall be capable of being reinitiated for the full duration within 2 USEC after the end of the suppression period.

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3.4.2.5.3 ISLS Conditions for Non-Suppression - The suppression of the decoder shall not take place when the received amplitude of the first interrogation pulse is 9 DB or more in excess of the received amplitude of the ISLS control pulse, or when the spacing of the first interrogation pulse and the ISLS is more than 2.6 USEC.

3.4.2.5.4 ISLS Gray Region - If the level of the interrogation pulse is greater than the level of the suppression pulse but greater by less than 9 DB, suppression may or may not take place.

3.4.2.5.5 ISLS Dynamic Range - The ISLS suppression and non-suppression characteristics shall apply over a received signal amplitude range from 3 DB above MTL to greater than 50 DB above that level.

3.4.2.6 Pulse Width Discrimination - Signals of received amplitude between MTL and 6 DB above this level, and of a duration less than 0.3 USEC on Modes 1, 2, 3/A and C or 0.2 USEC on Mode 4 at the antenna input, shall not cause the transponder to initiate more than 10 percent reply or suppression action. In the case of Mode 4, appropriate processing shall be performed on the challenge video pulses and compliance shall be determined using a simulated transponder-computer configuration. With the exception of pulses with amplitude variations approximating a normal interrogation or suppression pulse pair condition, any pulse or pair of pulses each of a duration more than 1.6 USEC shall not cause the transponder to initiate reply or suppression action.

#3.4.2.7 Transponder Delay - The transponder delay between the second pulse of a Mode 1, 2, 3/A, or C interrogation received at the antenna terminal and the first reply pulse delivered at the antenna terminal shall be between 2.5 and 3.5 USEC. The variation in delay between Modes 1, 2, 3/A, and C shall not exceed 0.2 USEC for nominally spaced interrogations. The transponder delay for Mode 4 measured from the radio frequency input of an interrogation fourth sync pulse to the radio frequency output of the first pulse in the first reply position, excluding KIT-1A/TSEC computer delay, shall be less than 3.0 USEC and greater than 2.0 USEC for all interrogation signal levels more than 3 DB above MTL.

3.4.2.8 Range Jitter - On any one of Modes 1, 2, 3/A, C or Test, the total jitter of the reply pulse code group, with respect to P_3, shall not exceed 0.1 USEC for receiver input levels between 3 and 50 DB above MTL.

Assumes computer delay time is referenced from the leading edge of the fourth sync pulse of the Mode 4 challenge video.

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3.4.2.9 Reply Characteristics - The transmitted replies as measured at the antenna terminal shall have the characteristics listed in table IX.

Table IX. Reply Pulse Characteristics

Characteristics	Value
Duration	0.45 ± 0.1 USEC
Rise time	0.05 to 0.1 USEC
Decay time	0.05 to 0.2 USEC
Amplitude Jitter	5 percent or less

3.4.2.9.1 Mode 1 Replies - For each pulse appearing at the Mode 1 decoder output, a pulse train shall be transmitted containing from none to a maximum of 5 information pulses plus 2 framing pulses which are spaced 20.3 ± 0.05 USEC. The information pulse spacing shall be in multiples of 2.9 USEC from the initial framing pulse with a maximum allowable deviation from this spacing of ± 0.05 USEC. The position where a sixth pulse would appear (17.4 ± 0.05 USEC from the initial framing pulse) will not be used. From the specified 5 information pulses, a total of 32 different codes remotely selectable are available.

3.4.2.9.2 Mode 2 Replies - For each pulse appearing at the Mode 2 decoder output, a pulse train shall be transmitted containing from none to a maximum of 13 information pulses plus 2 framing pulses which are spaced 20.3 ± 0.05 USEC. The information pulse spacing shall be in multiples of 1.45 USEC from the initial framing pulse with a maximum allowable deviation from this spacing of ± 0.05 USEC. The use of the seventh information pulse position (10.15 ± 0.05 USEC from the framing pulse) is covered in 3.4.2.14.5. Facility for selecting any of the 4096 codes available shall be provided on the equipment front panel.

3.4.2.9.3 Mode 3/A Replies - For each pulse appearing at the Mode 3/A decoder output, a pulse train shall be transmitted containing from none to a maximum of 13 information pulses plus 2 framing pulses which are spaced 20.3 ± 0.05 USEC. The information pulse spacing shall be in multiples of 1.45 USEC from the initial framing pulse with a maximum allowable deviation from this spacing of ± 0.05 USEC. The use of the seventh information pulse position (10.15 ± 0.05 USEC from the framing pulse) is covered in 3.4.2.14.5. From the specified 12 information pulses, a total of 4096 different codes remotely selectable are available.

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3.4.2.9.4 Test Mode Replies - For each pulse appearing at the Test Mode decoder output when the Test Mode is remotely enabled a pulse train shall be generated as described in Mode 3/A (3.4.2.9.3).

3.4.2.9.5 Mode 4 Replies - Mode 4 encoding is performed in an external computer.

3.4.2.9.6 Mode C Replies - For each pulse appearing at the Mode C decoder output, when the receiver is connected to an altitude computer-transducer, a pulse train shall be generated containing 1 or more information pulses, plus 2 framing pulses spaced 20.3 USEC and the Special Position Identification (SPI) pulse (3.4.2.10.) The information pulses shall be referenced from the first framing pulse in multiples of 1.45 USEC with a maximum allowable deviation from this spacing of ± 0.05 USEC. The position where a seventh pulse (X pulse) would appear (10.15 ± 0.05 USEC from the initial framing pulse) and where a ninth pulse would appear (13.05 ± 0.05 USEC from the initial framing pulse) will not be used. Whenever the thirteenth pulse position (18.85 ± 0.05 USEC) is chosen, the SPI pulse shall also be generated at the output. The SPI pulse shall be spaced 24.65 ± 0.05 USEC from the first framing pulse. Control leads shall be provided to accept digitized altitude information from the computer-transducer. In the absence of such a transducer, only bracket pulses shall be generated.

3.4.2.9.7 Emergency Replies, Modes 1, 2, and 3/A - When the receiver-transmitter is switched to Emergency, the coder shall generate multiple pulse trains as follows:

3.4.2.9.7.1 Mode 1 Emergency Replies - For each trigger pulse appearing at the Mode 1 decoder, the code in use will appear in the first train, following by 3 sets of framing pulses with these pulses appearing at 24.65 ± 0.1, 44.95 ± 0.15, 49.30 ± 0.20, 69.60 ± 0.25, 73.95 ± 0.30 and 94.25 ± 0.35 USEC.

3.4.2.9.7.2 Mode 2 Emergency Replies - For each trigger pulse appearing at the Mode 2 decoder output, the code in use will appear in the first train, followed by 3 sets of framing pulses, with the pulses spaced as for Mode 1.

3.4.2.9.7.3 Mode 3/A Emergency Replies - For each trigger pulse appearing at the Mode 3/A decoder output, the code number 7700 will appear in the first train, followed by 3 sets of framing pulses, with the pulses spaced as for Mode 1. Replies in Modes 1, 2, and 3/A shall be enabled regardless of the setting of the remote mode-select switches.

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3.4.2.9.7.4 Emergency Replies, Modes 4 and C - The emergency function does not affect Modes 4 and C.

3.4.2.10 Identification of Position Replies - Provision shall be made for IDENT operation. A time delay shall be provided so that when the IDENT control lead is initially grounded (regardless of duration of grounding period) by remote control, IDENT action will be initiated and will continue for a pre-set period between 15 and 30 seconds. The IDENT timing operation shall be capable of being reinitiated at any time. When the receiver-transmitter is switched to IDENT, the coder shall generate code replies with the following characteristics:

3.4.2.10.1 Mode 1 IDENT Replies - For each trigger appearing at the Mode 1 decoder output, the code in use shall appear twice. The second pulse train shall be spaced 24.65 ± 0.05 USEC as measured between the leading edges of the first framing pulses of the two trains.

3.4.2.10.2 Mode 2 IDENT Replies - For each trigger pulse appearing at the Mode 2 decoder output, the code in use shall appear followed by a single pulse (SPI pulse) spaced 24.65 ± 0.05 USEC from the first framing pulse.

3.4.2.10.3 Mode 3/A IDENT Replies - For each trigger pulse appearing at the Mode 3/A decoder output, the code in use shall appear followed by a single pulse (SPI pulse) spaced 24.65 ± 0.05 USEC from the first framing pulse.

3.4.2.10.4 IDENT Replies in Other Modes - The IDENT function shall not affect the normal operation of Mode C and Mode 4.

3.4.2.11 Mode C Altitude Reporting Provisions - When a Mode C interrogation is decoded, the coder shall generate a pulse train as described in 3.4.2.9.6. The information pulses are remotely selected by an altitude computer-digitizer.

3.4.2.11.1 Mode C Altitude-Code Control Characteristics - Input connections shall be provided on the receiver-transmitter to accept Mode C reply code information from devices using either electronic keying or a disc-brush encoder. Mode C control characteristics shall be as follows:

a. A turn-off (information bit not present) shall occur under either of the following conditions:

   (1) When a low-impedance voltage applied to the control lead is at least +18.5V but not exceeding +30V.

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(2) When an external resistance in excess of 100,000 ohms to applied to the control lead.

b. A turn-on (information bit present) shall occur under either of the following conditions:

(1) When a low impedance voltage applied to the control lead is +2.0V or less.

(2) When an external resistance of 3000 ohms or less to ground is applied to the control lead. This requirement applies to a single transponder operating alone and not paralleled with other transponders or remote altitude indicators.

c. Steady-state current in any control lead shall be between 1.25 to 2.5 MA when connected to ground.

3.4.2.11.2 Noise Filtering on Mode C Leads - Suitable filtering shall be provided within the receiver-transmitter to limit the capacitance discharge current from an external encoder disc brush to 1.5 MA or less in addition to the steady state brush current. The use of inductive-type filtering shall not be permitted.

3.4.2.11.3 Mode C Lead Source and Transient Voltages - The source of the energizing voltages for the coding circuit within the receiver-transmitter shall be such that the open circuit voltage on the control leads will not exceed 30 V. Transient protection on this source voltage shall be provided so that, irrespective of the transients which might appear on the input power leads to the receiver-transmitter, the peak inverse voltage on the control leads shall not exceed 30 V.

3.4.2.12 Auxiliary Trigger Operation - An input shall be made available with an input impedance of 90 ± 10 percent ohms for accepting external triggers with characteristics as listed in table X.

Table X. Auxiliary Trigger Pulse Characteristics

Characteristics	Value
Duration	0.30 to 1.5 USEC
Amplitude	15 to 30 V
Rise Time	0.1 USEC or less
Decay time	0.2 USEC or less

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Polarity	Positive
Pulse spacing	2.0 USEC minimum

For each trigger received the equipment shall transmit a single reply pulse with characteristics described in 3.4.2.9.

3.4.2.13 Standby Operation - In the STANDBY position of the master control on the Control Box C-6280( )(P)/APX, the equipment shall operate normally except as follows:

a. Replies to Mode 1, 2, 3/A, 4, C, Auxiliary Trigger, and Test Interrogations shall be inhibited.

b. The absence of replies in Mode 4 due to the Standby function shall enable the Mode 4 caution light in the same manner as if replies were absent due to other reasons (see 3.4.2.15.2.4).

c. Full functional capability shall be restored immediately on switching from STANDBY to NORMAL or LOW.

3.4.2.14 Code Number Assignments - Every configuration of Mode 1, 2, and 3/A information pulses shall be assigned a code number. The following numbers shall be used:

3.4.2.14.1 Mode 1 Codes - Code numbers 00(00) through 73(00) shall designate the Mode 1 codes. The applicable 5 information pulses shall be divided into 2 groups (A and B) with A containing the first 3 pulses and B containing the last 2 pulses. The information pulses in group A shall be numbered in order serially (1, 2, and 4) and the 2 pulses in group B shall be numbered in order (1 and 2). The appropriate code number shall be obtained by adding the pulse numbers (1, 2, and 4) of the pulses present within each group and combining the sums in order (A, B) to form the serial code numbers. Thus, code 03(00) consists of no pulses in group A, and pulses 1 and 2 of group B; code 51(00) consists of pulses 1 and 4 of group A, and pulse 1 of group B, etc.

3.4.2.14.2 Mode 2 Codes - Code numbers 0000 through 7777 shall designate the Mode 2 codes. The applicable 12 information pulses shall be divided into 4 groups of 3 pulses. The individual pulses of each group shall be spaced 2.9 USEC apart and numbered in order (1, 2, and 4). The position of the first pulse of each group, measured from the initial framing pulse, shall be as shown in table XI.

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Table XI. Mode 2 Pulse Positions

Group	Time From First Framing Pulse
A	02.9 USEC
B	11.6 USEC
C	01.45 USEC
D	13.05 USEC

The appropriate code number shall be determined by adding the sum (1, 2, and 4) of the pulses present within each group. These sums shall then be combined in order (A, B, C, D) to form a 4-digit serial number for the particular code. For example, code 2435 will have information pulses at the 1, 3, 4, 9, 12, and 13 information pulse positions.

3.4.2.14.3 Mode 3/A Codes - Code numbers 0000 through 7777 shall designate the Mode 3/A codes. The same system shall apply to Mode 3/A as that described in 3.4.2.14.2.

3.4.2.14.4 Mode C Codes - The hybrid reflected binary (gray) code shall designate the Mode C codes. The X and D_1 pulse positions shall not be used. The SPI and D_4 pulses shall be considered a single bit of information and both pulses shall be generated whenever D_4 is enabled.

3.4.2.14.5 X-Pulse Codes - A pulse spaced 10.15 ± 0.05 USEC from the initial framing pulse shall be designated as the "X" pulse. This pulse position is normally unused. Provisions shall be made such that, when a single external control lead is grounded, all replies in Modes 1, 2, and 3/A will include this pulse in adddition to the normal framing and information pulses.

3.4.2.15 Mode 4 Performance

3.4.2.15.1 Mode 4 Input Signals - The receiver-transmitter shall accept the following input and control signals:

3.4.2.15.1.1 Mode 4 Reply Input - A video input shall be provided for application of Mode 4 reply triggers with characteristics as listed in table XII.

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Table XII. Mode 4 Trigger Pulse Characteristics

Characteristic	Value
Pulse width	0.3 to 0.7 USEC
Rise time	0.1 USEC or less
Decay time	0.25 USEC or less
Polarity	Positive
Amplitude	3 to 5.0 V across impedance of 90 ± 10 percent ohms
Pulse spacing	In normal operation, Mode 4 replies consist of three pulses having a nominal spacing 1.8 ± 0.2 USEC between adjacent pulses. The total spacing between the first and third pulses will not differ from its nominal value of 3.6 USEC by more than 0.2 USEC.
Undesired signals plus quiescent DC voltage	Less than ± 0.5V
Negative overshoot caused by receiver-transmitter	Less than 1.5V

For each input trigger, an output pulse shall be generated having characteristics described in 3.4.2.9.

3.4.2.15.1.2 Disparity Input - A video input shall be provided for application of Mode 4 disparity trigger pulses to the equipment. Characteristics of the disparity pulse, when measured across 90 ± 10 percent ohms, shall be as shown in table XIII.

Table XIII. Disparity Pulse Characteristics

Characteristic	Value
Amplitude	3 to 5.0 V
Polarity	Positive
Duration	0.3 to 1.0 USEC
Rise time	0.15 USEC maximum

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Table XIII. Disparity Pulse Characteristics Cont'd

Characteristic	Value
Fall time	0.5 USEC maximum
Undesired signals plus quiescent DC Voltage	Less than ± 0.5V
Negative overshoot cause by load	Less than 1.5V
Timing from fourth sync pulse	Within 200 USEC

3.4.2.15.1.3 Audio Enable - A control lead shall be provided to permit aural montoring (3.4.2.15.2.5) of the Mode 4 interrogating signal when the audio enable control lead is grounded.

3.4.2.15.2 Mode 4 Output Signals - The receiver-transmitter shall furnish the following output signals:

3.4.2.15.2.1 Mode 4 Enable Trigger - The equipment shall generate a Mode 4 enabling trigger for an external coder-decoder upon recognition of the Mode 4 sync pulses described in 3.3.13. The Mode 4 trigger pulse shall have characteristics as listed in table XIV.

Table XIV. Mode 4 Enable Trigger Pulse Characteristics

Characteristic	Value
Amplitude	1.5 to 5.0V across an impedance of 90 ± 10 percent
Rise time	0.1 USEC or less
Decay time	1.0 USEC or less
Duration	0.5 to 3.0 USEC
Polarity	Positive

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Delay	Spacing from the nominal position of the fourth sync pulse of the challenge video output will be from + 0.2 to + 0.6 USEC, nominally. This spacing will not vary more than ± 0.07 USEC throughout all extreme operating conditions or combinations thereof.
Undesired signals plus quiescent DC voltage	Less than ± 0.5V

3.4.2.15.2.2 Mode 4 Challenge Video - The equipment shall route interrogation signals to an external coder-decoder. The output pulses shall have the same characteristics as those specified 3.3.13 for Mode 4, plus those shown in table XV.

Table XV. Mode 4 Challenge Video Pulse Characteristics
(additional to Mode 4 pulse requirements)

Characteristic	Value
Duration	0.4 to 0.6 USEC
Amplitude	1.5 to 5.0V across 90 ± 10 percent ohms
Polarity	Positive
Undesired signals plus quiescent DC voltage	Less than ± 0.5 V
Rise time	0.1 USEC maximum
Decay time	0.2 USEC maximum
Pulse Train droop	15 percent maximum

3.4.2.15.2.3 Mode 4 Reply Light Enable - Provision shall be made to enable the Mode 4 reply light only when four or more Mode 4 replies are transmitted within 0.033 second interval, or when replies are transmitted at a rate of 50 or more per second for 0.25 seconds, as indicated by the presence of Mode 4 reply inputs followed within 0.5 USEC by the transmission of RF reply pulses.

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The reply light enabling voltage shall not be generated if transmitted replies are generated at a constant rate of less than ten per second. The reply light shall remain enabled for a period of 2 to 5 seconds after the end of the enabling conditions.

3.4.2.15.2.4 Mode 4 Caution Light Enable - Provisions shall be made to enable an external caution light whenever either of the following conditions exist:

a. Zeroized Mode 4 reply code or computer failure is indicated by grounding of a control lead by the external Mode 4 computer.

b. The receiver-transmitter has failed to reply to valid Mode 4 interrogations four or more times within an 0.033 second interval or to valid Mode 4 interrogations at a rate greater than 50 pulses per second for 0.25 seconds. A failure shall be considered to have occurred when a Mode 4 sync pattern is recognized but not followed within 270 USEC by either a Mode 4 reply indication (3.4.2.15.2.3), or a disparity pulse (3.4.2.15.1.2). An exception to the "failure to reply" indication shall apply when the failure to reply to a valid Mode 4 interrogation occurs at a constant rate of less than 10 times per second. The lamp shall remain on for a period of from 2 to 5 seconds after the end of the enabling conditions.

3.4.2.15.2.5 Mode 4 Audio Output/Enable - An audio signal shall be available between the audio output leads whenever both of the following conditons are satisfied:

a. An external control lead designated Audio Enable is grounded.

b. Presence of a Mode 4 trigger (3.4.2.15.2.1) when not followed by a disparity pulse (3.4.2.15.1.2).

3.4.2.15.2.6 Mode 4 Audio Output Signal Characteristics - The audio return lead shall not be grounded in the receiver-transmitter. The audio signal shall consist of pulses having characteristics as listed in table XVI.

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Table XVI. Audio Output Signal Characteristics

Characteristic	Value
Amplitude	0 to 3V (adjustable) into 150 ohms resistive load. 0 to 5V (adjustable) into 600 ohms resistive load.
Frequency	Repetition frequency of Mode 4 interrogation trains for interrogation rates up to 1,000 per second. Above 1,000 interrogations per second, the pulse frequency output shall be greater than 500 per second and less than or equal to the interrogation rate.
Pulse width	At least 500 USEC for interrogation rates less than 200 per second.

3.4.3 Receiver Detailed Performance Requirements - The receiver shall be of the tuned radio-frequency crystal video type incorporating at least three stages of radio frequency amplification prior to the detector.

3.4.3.1 Receiver Frequency - The receiver shall operate at a fixed frequency of 1030 MHz.

3.4.3.2 Receiver Frequency Stability - Under all combinations of service conditions specified herein, the frequency drift from that measured under conditions of 3.3.8 shall not exceed ± 1.5 MHz.

*3.4.3.3 Receiver Noise Figure - The receiver noise figure shall not exceed 13 DB.

3.4.3.4 Receiver Dynamic Range - Performance requirements specified in 3.4.3 and subparagraphs shall be met with interrogation signals from minimum triggering level to at least 50 DB above that level unless otherwise specified.

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3.4.3.5 Receiver Bandwidth - Overall receiver bandwidth shall be in accordance with table XVII. The bandpass shall be essentially gaussian and shall accommodate interrogation signal frequencies of 1030 ±2.5 MHz with not more than 3 DB variation in sensitivity.

Table XVII. Overall Receiver Bandwidth

-6 DB points	greater than 7.0 MHz
-60 DB points	less than ±25 MHz

3.4.3.6 Receiver RF Rejection - The CW and pulse responses shall be 60 DB less than that measured at 1030 MHz for all frequencies less than 1005 MHz and greater than 1055 MHz.

3.4.4 Decoder Detailed Performance Requirements - The equipment shall incorporate decoding circuits for the purpose of distinguishing and causing replies to the six modes of interrogation described in 3.3.13. In addition, interrogations made up of pulses of 0.9 to 1.2 USEC width with rise times of 0.2 USEC and decay time of 0.3 USEC, shall be decoded and replies generated with no degradation in performance, except that the decoder tolerance may be proportional to pulse width. This circuitry shall be such that the reply transmitted by the equipment shall always correspond to the mode of interrogation. Control, Transponder Set C-6280 ( ) (P)/APX provides discrete control of the decoding of Modes 1, 2, 3/A, C, and Test.

3.4.4.1 Decoder Tolerance, Mode 1 - Interrogation pulses spaced 2.8 to 3.2 USEC shall cause the equipment to generate Mode 1 replies. Pulses spaced less than 2.5 or greater than 3.8 USEC shall not cause such replies.

3.4.4.2 Decoder Tolerance, Mode 2 - Interrogation pulses spaced 4.8 to 5.2 USEC shall cause the equipment to generate Mode 2 replies. Pulses spaced less than 4.2 or greater than 5.8 USEC shall not cause such replies.

3.4.4.3 Decoder Tolerance, Mode 3/A - Interrogation pulses spaced 7.8 to 8.2 USEC shall cause the equipment to generate Mode 3/A replies. Pulses spaced less than 7.2 or greater than 8.8 USEC shall not cause such replies (except see 3.4.4.6).

3.4.4.4 Decoder Tolerance, Mode 4 - Four-pulse interrogation signals with pulses spaced within ±0.1 USEC from nominal positions shall cause the equipment to generate Mode 4 enable triggers. Pulses spaced greater than ±0.6 USEC from nominal positions shall not cause such triggers.

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3.4.4.5 Decoder Tolerance, Mode C - Interrogation pulses spaced 20.8 to 21.2 USEC shall cause the equipment to generate Mode C replies. Pulses spaced less than 20.2 or greater than 21.8 USEC shall not cause such replies.

3.4.4.6 Decoder Tolerance, Test Mode - Interrogation pulses spaced 6.3 to 6.7 USEC shall cause the equipment to generate Mode 3/A replies. Pulses spaced less than 5.7 or greater than 7.3 USEC shall not cause such replies (except see 3.4.4.3).

3.4.5 Transmitter Detailed Performance Requirements - The equipment shall respond to interrogations defined in 3.3.13 by transmitting pulse-train replies for each interrogation received, depending on the mode of interrogation and upon equipment control settings selected. The transmitter shall consist of a grid-modulated oscillator and an RF driven power amplifier.

3.4.5.1 Transmitter Frequency - The transmitter shall operate at a fixed frequency of 1090 MHz.

3.4.5.2 Transmitter Frequency Stability - Any reasonable combination of service conditions shall not cause the transmitter frequency to drift more than ±3.0 MHz from 1090 MHz measured under the standard conditions of 3.3.8.

3.4.5.3 Transmitter Power Output - The transmitter shall deliver 27 ±3 DB above one watt to the antenna terminal of the equipment under all reasonable combinations of specified conditions. When the equipment is delivered, the transmitter power output shall be at least 27DB above one watt when measured under the conditions of 3.3.8 and at a 1 percent duty cycle.

3.4.5.3.1 Transmitter Power Output Variation with Input Voltage - When the power supply is operating with the direct-current input, transmitter peak power measured under the nominal conditions of 3.3.8 shall not be reduced by more than 3.0 DB when the input supply voltage is reduced to 17.0 VDC.

3.4.5.3.2 Transmitter Power Output Variation Within A Pulse Train - The peak power of any pulse in a group of pulses specified in 3.4.2.9 shall not differ from the power of any other pulse in that group by more than 1.0 DB.

*3.4.5.4 Transmitter Replacement - The equipment design shall permit the replacement of the transmitter without the necessity of altering equipment adjustments other than oscillator frequency, power amplifier tuning, and pulse width adjustment.

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3.4.6 Power Supply Detailed Performance Requirements - An integral power supply shall be provided to accept input power as described in 3.3.12 and provide proper conversion to the voltages and currents required by the remainder of the equipment. This supply shall be of solid state, modular, quick-replaceable design and shall incorporate to the maximum extent possible nonreparable, easily-replaceable subassemblies to facilitate maintenance by the substitution methods.

3.4.6.1 Power Supply Reserve Capability - To allow for future system changes and expansion, the power supply shall be capable of supplying the additional current drains as listed in Table XVIII beyond the maximum required for the equipment described herein. These additional current drains shall not degrade performance beyond that specified in 3.4.

Table XVIII. Power Supply Reserve Capability

Voltage	Current
+25 VDC	10 milliamperes (MA)
+12 VDC	50 MA
+6 VDC	20 MA
-6 VDC	10 MA
6.3 V (filament)	100 MA

3.4.6.2 Power Supply Protection - The power supply shall be self-protecting against faults which place an overload (above reserve capability) or short circuit on any or all of the power supply outputs. The power supply need not operate during the presence of such faults, but the use of fuses for this protection will not be allowed, and operation shall return to normal once the fault has cleared.

3.5 Detail Requirements

3.5.1 Receiver-Transmitter, Radio RT-859 ( )/APX-72 - The receiver-transmitter shall fulfill the following requirements:

3.5.1.1 Functional Requirement - The receiver portion of this equipment shall receive RF energy from the antenna, amplify and decode it and initiate the proper transmitted reply pulses. The transmitter shall generate RF power and transmit amplified pulse replies to the load specified herein.

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*3.5.1.2 Form Factor - The equipment shall be enclosed in a two-piece, diecast aluminum case which shall also serve as the chassis. Overall dimensions, excluding the mount, shall not exceed 6 3/8 inches wide by 6 1/2 inches high by 13 inches long.

*3.5.1.3 Weight - The weight of the receiver-transmitter shall not exceed 15.05 pounds. The weight of the mount specified in 3.5.3.1 shall not exceed 1.45 pounds. The weight of the mount specified in 3.5.3.2 shall not exceed 2.1 pounds.

3.5.1.4 Contents - The receiver-transmitter shall consist of the following assemblies.

*3.5.1.4.1 RF Assembly - This unit shall consist of the lower part of the two piece case and shall contain the low-pass filter, diplexer, receiver preselector filter and RF amplifier, detector, video amplifier, transmitter modulator, oscillator, RF power amplifier, sensitivity board and connector and filters. The antenna connector and pressure valve shall be located on the front of this assembly.

3.5.1.4.2 Front-panel Control and Connector Assembly - This unit shall contain primary supply voltage fuses, Mode 2 code control switches, power and control connector, filters and a handle. Space shall be reserved for the future mounting of a ITT Cannon Electric (or equivalent) Type DBM connector and its associated conductors.

3.5.1.4.3 Upper Chassis - This unit shall consist of the upper part of the two-piece case and shall contain the necessary guides, etc. for inserting the assemblies listed in 3.5.1.4.3.1 through 3.5.1.4.3.7 and 3.5.1.4.3.9 herein. A pressurized connector with at least 91 contacts shall be used to provide connections between the pressurized compartment and the front panel.

3.5.1.4.3.1 Processor Plug-In Board - This unit shall provide the processing necessary to meet specified requirements.

3.5.1.4.3.2 Decoder Plug-In Board - This unit shall provide decoding, sensitivity-reduction and suppression functions.

3.5.1.4.3.3 Mode 4 Function Plug-In Board - This unit shall provide all unique functions necessary for Mode 4 operation. Performance on Modes 1, 2, 3/A, Test and C shall not be affected by the absence of this board from the equipment.

3.5.1.4.3.4 Encoder Clock Plug-In Board - This unit shall provide the timing necessary for encoding purposes.

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3.5.1.4.3.5 Encoder Control Plug-In Board - This unit shall provide code configuration control functions.

3.5.1.4.3.6 Encoder Gating Plug-In Board - This unit shall provide gating functions necessary for encoding.

3.5.1.4.3.7 Delay-Line Plug-In Board - This unit shall contain the delay-line necessary for decoding.

3.5.1.4.3.8 Provision for Spare Plug-In Board - Guides, chassis-cutout and space shall be provided for future employment of an additional plug-in board identical in form-factor to items 3.5.1.4.3.1 through 3.5.1.4.3.7. Board and connector are not to be provided.

3.5.1.4.3.9 Power-supply Assembly - This unit shall provide the required regulated low and high voltages.

3.5.1.5 Controls

3.5.1.5.1 Primary Power Controls - The equipment shall incorporate control circuitry to accomplish the following when used with Control, Transponder Set C-6280 ( ) (P)/APX in accordance with MIL-C-38257:

a. In the "Standby" position of the control, the filament voltage as well as all supply voltages shall be rated value, and the modulator shall be disabled.

b. In the "Normal" position of the control, all voltages shall be applied, the modulator shall be enabled and the unit shall be capable of normal operation immediately.

3.5.1.5.2 Mode 2 Controls - Thumbwheel switches shall be accessible on the Front-panel Control and Connector Assembly to select Mode 2 coding.

3.5.1.5.3 Control Lead Length - All control functions in the equipment shall operate satisfactorily when the interconnecting cable between the equipment and the control box (3.5.2) is up to 50 feet in length.

*3.5.1.6 Electrical Connection - Connections to external circuits shall be provided as shown in table XIX. Receptacle J1 shall be secured to the transponder with tapped inserts on the front panel to lock the airframe mating plug to the transponder. The tapped insert shall be keyed so as to assure permanent attachment to the case.

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Table XIX. Electrical Connections

Reference Designation	Receptable Type	Function
J1	ITT Cannon Electric Part No. DPJ-59C10-33P-B-2 (or equivalent)	Power, control and video connector
J5	Pressurized "N" type female connecotr	Antenna connector

3.5.1.6.1 Pin Connections - The signals and control functions listed in table XX shall appear at the designated pins of the power, control and video connector.

Table XX. Pin Connections

Pin Number	Function
1	AC return
2	+ 28 VDC input
3	System ground
4	Test mode control
5	115 VAC input
6	Switched 28 VDC output power
7	Switched 115 VAC output power
8	Mode C, A_1 control
9	Audio output
10	Audio common
11	Suppression output
12	Suppression input

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Table XX. Pin Connections

Pin Number	Function
13	Auxiliary trigger
14	Sensitivity control
15	Mode 2 enable
16	Standby control
17	Mode C, B_1 control
18	Mode 3/A enable
19	Emergency control
20	Power relay control
21	IDENT control
22	Mode 1 enable
23	Mode C, B_2 control
24	Mode 3/A, B_4 control
25	Mode 3/A, B_2 control
26	Mode 3/A, B_1 control
27	Mode 3/A, A_4 control
28	Mode 3/A, A_2 control
29	Mode 3/A, A_1 control
30	Mode C, B_4 control
31	Mode C, C_1 control
32	Mode C, C_2 control
33	X pulse control

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Table XX. Pin Connections (Cont'd)

Pin Number	Function
34	Reply light enable
35	Mode C, C_4 enable
36	Mode 1, B_2 enable
37	Mode 1, B_1 enable
38	Mode 1, A_4 enable
39	Mode 1, A_2 enable
40	Mode 1, A_1 enable
41	Mode C, D_2 enable
42	Mode C, D_4 enable
43	Mode 3/A, C_1 enable
44	Mode 3/A, C_2 enable
45	Mode 4 challenge video
46	Mode 4 enable trigger output
47	Mode 4 reply trigger input
48	Mode 4 disparity trigger input
49	DC power supply operation
50	Mode 3/A, C_4 control
51	Audio enable
52	Mode 4 zeroize code indication
53	+28 VDC caution light enable
54	Mode C, A_2 control

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Table XX. Pin Connections (Cont'd)

Pin Number	Function
55	Mode C, A_4 control
56	Mode 3/A, D_1 control
57	Mode 3/A, D_2 control
58	Mode 3/A, D_4 control
59	Mode C enable

3.5.1.7 Printed Circuit Boards - Any board which requires electrical continuity from one side of the board to the other shall employ plated through holes to assure reliability and reparability. Also, any plug-in contacts shall be gold plated to a thickness of not less than 0.08 mils.

3.5.1.8 Heat Transfer Compound - An appropriate thermal compound approved by the procuring agency shall be used in clamping the power supply module, and receiver radio frequency (RF) amplifier and the transmitter cavity(ies) to the case. This compound shall be used in such a way as to assure efficient heat transfer to the case, and appropriate instructions concerning its use when components are replaced shall be included in the maintenance manual.

3.5.1.9 Pressurization - With the exception of the transponder front panel and control and connector assembly, the equipment shall be pressurized in accordance with 3.2.18 of MIL-E-5400 to a pressure of 5 pounds per square inch, guage (PSIG) at sea level.

3.5.1.9.1 Non-pressurized Operation - The equipment shall meet electrical performance requirements up to an altitude of 30,000 feet without pressurization.

3.5.1.10 Adjustments Prior to Delivery - Before the equipment is delivered by the contractor, the transmitter frequency shall be adjusted to 1090 ± 0.5 (MHz) when the input impedance presented to the equipment is such that a SWR of not more than 1.1 to 1 exists. The center of the receiver pass band (-6 DB points) shall be adjusted to a frequency of 1030 ± 0.5 MHz.

3.5.2 Control, Transponder Set C-6280 ( ) (P)/APX - The C-6280 ( ) (P)/APX-72 is not to be furnished as part of this equipment.

3.5.3 Mountings - A mount shall be provided. This shall be either a hardmount (3.5.3.1) or an isolation-mount (3.5.3.2) as required by the contract.

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*3.5.3.1 Mounting MT-3809 ( )/APX-72 - Unless otherwise specified in the contract, a hard-mount capable of meeting vibration and shock requirements of 3.3.9 shall be provided. Two front-located, self-locking screw fasteners shall allow convenient removal of the receiver-transmitter from the mounting. Overall dimension of the receiver-transmitter, complete with mounting, shall not exceed 6 3/8 inches width by 6 1/2 inches height by 13 5/8 inches length.

*3.5.3.2 Mounting MT-3948 ( )/APX-72 - When specifically called for in the contract, an isolation mount capable of meeting the vibration requirements of (3.3.9.) shall be provided. The overall dimensions of the receiver-transmitter, complete with mount, shall not exceed 8 1/4 by 6 11/32 by 13 5/8 inches, including sway space. There shall be 4 mounting centers spaced 3 1/2 by 9 1/16 inches, and each mounting center shall require a square pattern of 4 holes, each 0.201 inches in diameter, on 1 3/8 inch centers.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

*4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

a. Preproduction (First Article) Tests

b. Initial Production Tests

c. Acceptance Tests

d. Life Tests

4.2 Preproduction (First Article) Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the approved test procedure of 4.6. The Government inspector and the procuring activity shall be advised when tests are to be conducted so that a Government representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

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4.2.1 Preproduction (First Article) Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review.

4.2.2 Scope of Preproduction (First Article) Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification, other applicable specifications, and the contract. Preproduction tests shall include environmental tests in accordance with the procedures of MIL-T-5422 and interference tests in accordance with MIL-I-6181, and MIL-E-55301. These tests shall also include reliability qualification tests (see 4.4.3).

4.2.3 Preproduction (First Article) Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests - One of the first ten production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. No other test shall be conducted on the equipment prior to starting the initial production tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Initial Production Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction workmanship, performance, and quality and that it meets all applicable requirements.

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4.3.2 Accessory Material - In addition to the complete equipment submitted for initial production tests the contractor shall also submit such accessory material and data necessary to test the equipment.

4.3.3 Initial Production Sample Approval - Approval of the initial production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material, or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the initial production sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government inspector. Contractors not having adequate facilities for conducting all required tests shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

a. Individual tests

b. Sampling tests

c. Reliability assurance tests

d. Special tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

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a. Examination of product

b. Operational test

c. Manufacturing run-in test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

*4.4.1.3 Manufacturing Run-In Test - Each equipment shall be operated under the conditions specified in table XXI for a period of 10 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Table XXI. Environmental Condition for Manufacturing Run-in Test

Environment	Value
Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 Hz (excluding resonant points) and a minimum amplitude of ± 3 g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 10 hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to the plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 10 hour period of operation following the 10 minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10 minute vibration period need not be repeated when it is certain the failure was not a result of the vibration.

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Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 10 hour period specified above may be composed of 5 hour periods to conform with standard working hours. This test is not required if a reliability assurance test is selected which includes a test on each equipment which consumes at least 10 hours of operation.

4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the Government inspector in accordance with table XXII.

Table XXII. Sampling Test Equipment Quantities

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Sampling Test
First 10	0 #
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

Sampling tests are not required when reliability production acceptance phase tests are conducted.

4.4.2.1 Scope of Sampling Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

a. Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

b. Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual and special tests.

#If by contract action the Initial Production Test (4.3) is deleted a sample test shall also be conducted on one equipment from the first 10 produced.

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c. Manufacturing run in test specified in 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.4.3 Reliability Assurance Tests - Reliability assurance tests shall be conducted using MIL-STD-781. Tests as required by both the qualification phase and the production acceptance (sampling) phase shall be conducted. Equipment selected for reliability assurance tests shall have first passed the individual tests.

4.4.3.1 Qualification Phase - Prior to the acceptance of equipments under the contract or order, a minimum of three (3) equipments shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests". The maximum number of equipments to be used shall be those listed in table 5 of MIL-STD-781. For the qualification phase, test level F shall be used. The accept-reject criteria for test plan I shall be used.

*4.4.3.2 Reliability Production Acceptance (Sampling) Phase Tests - The equipment, throughout production, shall be tested as outlined in MIL-STD-781 (as modified herein) under the section entitled "Production Acceptance (Sampling) Phase of Production Reliability Tests". Test Level F shall be used.

*4.4.3.2.1 All Equipment Test - Each equipment produced except those submitted for the Reliability Qualification Test, shall be tested for 24 hours. Prior to the 24 hour test on each equipment, a burn-in period may be used at the option of the contractor. If the burn-in period is to be used the details thereof must be included in the approved test procedures. To determine whether the MTBF is being met at any time during the contract the operating test hours and the failures thereon (not counting burn-in failures or burn-in operating time) shall be totaled and the results compared with the reject line of Test Plan II of MIL-STD-781. (Extend the line as necessary to accomodate the data.) These totals shall accumulate so that at any one time the experience from the beginning of the contract is included. At the conclusion of each month the test results shall be sent to the procuring activity and to the Naval Air Systems Command, Attention: Avionics Division. At any time that the current totals of test hours and test failures plotted on Test Plan II curves show a reject situation, the procuring activity shall be notified. The procuring activity reserves the right to stop the acceptance of equipment at any time that a reject situation exists pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment workmanship, etc., so that the entire compilation will show other than a reject decision.

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4.4.3.3 Reliability Assurance Test Details - The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the test, etc., shall be part of the test procedures to be submitted and approved by the procuring activity prior to the beginning of the qualification test phase of the reliability assurance tests.

4.4.3.4 Special Failure Criteria - Failures shall not be counted as relevant failures for reliability acceptance for the following components after the specified hours of test are accumulated on each component.

a. Transmitter - 500 hours.

4.4.4 Special Tests - Special tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.4.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

a. On an early equipment after an engineering or material change.

b. Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity)

4.4.4.2 Scope of Special Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

a. Determine the cause of failure.

b. Determine if the failure is an isolated case or design defect.

c. Submit to the procuring activity for approval proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

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d. Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

*4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life tests. The test shall be of 300 hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the Government inspector in accordance with the following. (Equipments which have successfully passed the initial production test, sampling tests, reliability tests, or special tests may be selected for life tests). When reliability tests are conducted, the life test shall be omitted if, during the reliability tests, a quantity of equipments equal to, or more than, that listed below receive at least 300 hours each of test time.

Table XXIII. Life Test Equipment Quantities

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the simulated service conditions listed in table XXIV.

Table XXIV. Life Test Conditions

Temperature	Normal room
Altitude	Normal ground (0-5000 feet)
Humidity	Room ambient
AC Voltage	115 ± 5V (at lowest applicable frequency)
DC Voltage	27.5 ± 2.0 V

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4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test.

a. Part type number

b. The circuit reference symbol

c. The part function

d. Name of the manufacturer

e. Nature of the failure

f. The number of hours which the part operated prior to failure

4.5.4.1 Failure Report - In the event of a failure, the Government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the Government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

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4.7 Reconditioning of Tested Equipment - Equipment which has been subjected to acceptance and life tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.8 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government inspector.

5. PREPARATION FOR DELIVERY

*5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555 and MIL-STD-794. In the event the equipment is not covered in MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

6. NOTES

6.1 Intended Use - This equipment is intended for universal utilization in rotary and fixed-wing aircraft which provide a Class 2 (or less) environmental in this application it furnished aircraft identification and altitude reporting features to ground or air interrogations. The equipment is also intended for use on surface ships and submarines for purposes of furnishing ship identification to ground, surface or air interrogations.

6.2 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

a. Title, number, and date of this specification

b. Selection of applicable levels of packaging and packing (see 5.1).

6.3 Precedence Of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence will apply:

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a. Contract - The contract will have precedence over any specification

b. This Specification - This specification will have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, will be specifically approved in writing by the procuring activity.

c. Referenced Specifications - Any referenced specification will have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications will apply to the extent specified.

6.4 Performance Objectives - Minimum size and weight, simplicity of operation ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which will be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval will be submitted to the procuring activity for consideration. Each request will be accompanied by complete supporting information.

6.5 Nonreparable Subassemblies - As a general rule nonreparable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life the sub-assembly is greater than that of a similar reparable one. With few exceptions (such as high voltage power supplies), the nonreparable subassembly should evidence a Mean-Time-to-Failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.6 Type Designations - The parentheses ( ), when used in the equipment type designation, will be deleted or replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.7. The complete type number will be used on nameplates, shipping records and instruction books, as applicable.

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6.7 Revisions - In specification revisions and superseding amendments an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

6.8 Associated Equipment - The equipment will operate with associated equipment listed in table XXV.

Table XXV. Associated Equipment

Item	Type Designation
Control	Control, Transponder Set, C-6280 ( ) (P)/APX
Antenna	Antenna, AS-133 ( )/APX, AT-741( )/APX and other equivalent antennas
Transponder-Computer	KIT-1A/TSEC
Pressure-Altitude Digitizer	CPU-66 ( )/A or equivalent
Test Set	Test Set, Transponder Set TS-1843 ( )/APX
Switching Unit	Switching Unit Antenna SA-1474 ( )/A or equivalent

6.9 Definitions -

6.9.1 Duty Cycle - Per cent duty cycle is defined by the following equation:

$$DC = \frac{PW \times RC \times NP}{10^4}$$

where:

DC is percent duty cycle
PW is pulse width in microseconds
RC is reply count in number of counts
NP is number of pulses

Preparing Activity
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Project No. 5821-N061

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-R-81435A(AS) RECEIVER-TRANSMITTER, RADIO RT-859( )/APX-72

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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